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Professor Roy S. Smith
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Subject: Corrections to the Least Squares Regression Lecture Transcript

Dear Professor Smith,

I hope this message finds you well. I have been reviewing the transcript of your lecture on least squares regression and have identified a number of errors — some arising from voice-recognition transcription artefacts, and others appearing to be factual inaccuracies when checked against standard historical sources. I have compiled them below for your information.

I should note that the corrections to historical details were verified against the Wikipedia articles for Carl Friedrich Gauss, Ceres (dwarf planet), and Franz Xaver von Zach. I recognise that these are secondary sources and that you may have access to more authoritative references; I offer the corrections in good faith and welcome any further clarification.

Voice-Recognition Transcription Errors

The following appear to be artefacts of automatic speech recognition, where words or names were misheard:

As transcribed	Corrected to
<i>right essential, right essential angle</i>	right ascension angle
<i>xeris</i>	Ceres
<i>over what happened to 44 days</i>	over what amounted to 44 days
<i>1805 genre published a paper sort of on the method of the method of</i>	1805 Legendre published a paper sort of on the method of
<i>conic sectors</i> (three instances)	conic sections
<i>con acceptors</i>	conic sections
<i>conic sections, conic sectors</i> (stray doublet)	conic sections
<i>matrix calculus, calculus.org</i>	matrixcalculus.org
<i>from et cetera</i>	from ETH

Attribution Error

The transcript read: “*within half a degree of where Zach said it would be in that but Zach, but Gauss, would not say.*”

This appeared to be a garbled version of the intended meaning. I corrected it to: “*within half a degree of where Gauss said it would be, but Gauss would not say,*” since it was Gauss who made the prediction, not Zach (Zach was the astronomer who published the observational data and subsequently recovered Ceres).

Expansion of First Mention of Zach

The first reference to “Zach” gave no indication of who he was. I added a brief identification: “Zach — Franz Xaver von Zach, editor of the *Monatliche Correspondenz zur Beförderung der Erd- und Himmels-Kunde* (Monthly Correspondence for the Promotion of Earth and Sky Science). He published the observational data,” so that readers unfamiliar with the history have sufficient context.

Historical Factual Corrections

The following corrections were made on the basis of historical sources:

As transcribed	Corrected to
<i>Titus Bode law</i>	Titius–Bode law (after Johann Daniel Titius, not “Titus”)
<i>22 observations / And he made 22 of them</i>	24 observations / And he made 24 of them (Piazzi made 24 observations of Ceres before losing it)
<i>seventh of December</i>	31st of December (Ceres was recovered by Zach on 31 December 1801; he also observed on 7 December but the confirmed recovery is attributed to the 31st)
<i>five years earlier</i>	ten years earlier (Gauss claimed to have been using the method since 1794 or 1795, approximately ten years before Legendre published in 1805)

I hope these corrections are useful. Please do not hesitate to let me know if any of my changes deviate from your original intent, or if you have a preferred phrasing for any of the passages.

Thank you for an engaging and illuminating lecture.

Yours sincerely,

David Ewing
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